

Applied Linear Algebra

Norm and Distance

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August 2026

The *Euclidean norm* of an n -vector $\mathbf{x}$, denoted $\|\mathbf{x}\|$, is the squareroot of the sum of the squares of its elements,

$$\begin{aligned}\|\mathbf{x}\| &= \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \\ &= \sqrt{\mathbf{x}^T \mathbf{x}} \\ &= \|\mathbf{x}\|_2\end{aligned}$$

This is also called as the *magnitude* of a vector.

We will avoid calling it the *length* of a vector.

$$\|\mathbf{0}\| = 0$$

$$\|\mathbf{e}_i\| = 1$$

$$\|[c]\| = |c|$$

$$\|\mathbf{1}_n\| = \sqrt{n}$$

we use terms such as a *small* norm and *large* norm when we want to compare the magnitudes of vectors.

- Nonnegative homogeneity. $\|\beta \mathbf{x}\| = |\beta| \|\mathbf{x}\|$
- Nonnegativity. $\|\mathbf{x}\| \geq 0$
- Definiteness. $\|\mathbf{x}\| = 0$ only if $\mathbf{x} = \mathbf{0}$.
- Triangle inequality. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

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Quiz:

Example for triangle inequality?

a norm is a function from a real or complex vector space to the non-negative real numbers that behaves in certain ways like the distance from the origin: it commutes with scaling, obeys a form of the triangle inequality, and is zero only at the origin.

Definition

General Norm A *general norm* is any real-valued function of an n -vector that satisfies the four properties listed above.

Two examples of general norms:

$$\textbf{1-norm} \quad \|x\|_1 = |x_1| + \cdots + |x_n|$$

$$\textbf{Infinity-norm} \quad \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$$

Infinity norm $\|\cdot\|_\infty$ is extensively used in ML-DSA, NIST's PQC signature scheme.

The norm is related to the root-mean-square (RMS) value of an n -vector $\mathbf{x}$:

$$\begin{aligned}\text{rms}(\mathbf{x}) &= \sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}} \\ &= \frac{\|\mathbf{x}\|}{\sqrt{n}}\end{aligned}$$

The RMS value of a vector $\mathbf{x}$ is useful when comparing norms of vectors with different dimensions.

If all the entries of a vector are close to α , then the RMS value of the vector is $|\alpha|$.

The RMS value tells us what a 'typical' value of $|x_i|$ is.

The norm of $\mathbf{1}_n$, the n -vector of all ones, is $\sqrt{n}$, but its RMS value is 1, *independent* of n .

A couple of *norm identities*

$$\begin{aligned}\|\mathbf{x} + \mathbf{y}\| &= \sqrt{\|\mathbf{x}\|^2 + 2\mathbf{x}^\top\mathbf{y} + \|\mathbf{y}\|^2} \\ (\mathbf{x} + \mathbf{y})^\top(\mathbf{x} - \mathbf{y}) &= \|\mathbf{x}\|^2 - \|\mathbf{y}\|^2\end{aligned}$$

- 1805 — Adrien-Marie Legendre. His criterion was to make the sum of the squares of the errors as small as possible.
- Karl Pearson is generally credited with introducing the term “root-mean-square” in 1895, although the underlying mathematical quantity, and even root-mean-square deviation, predates the terminology.
- Pearson’s important contribution was to name and formalize terminology around it.
- In his 1893 lectures, he introduced *standard deviation* as a replacement for the older *root mean square* error.

If $\mathbf{a}$ and $\mathbf{b}$ are n -vectors,

$$\text{dist}(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$$

The RMS value of the vector difference, *RMS deviation* between vectors $\mathbf{a}$ and $\mathbf{b}$ is

$$\begin{aligned}\text{RMSD} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - b_i)^2} \\ &= \frac{\text{dist}(\mathbf{a}, \mathbf{b})}{\sqrt{n}}\end{aligned}$$

Euclidean Distance - Example

Given three vectors

$$\mathbf{u} = [1.5, -2.5, 1.5, -1.5]$$

$$\mathbf{v} = [0.5, -1.5, 3.5, -0.5]$$

$$\mathbf{w} = [-1.5, 1.5, 2.5, -2.5]$$

And the formula

$$\text{dist}(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + (x_4 - y_4)^2}$$

$$\mathbf{u} - \mathbf{v} = [1.0, -1.0, -2.0, -1.0]$$

$$\begin{aligned}\text{dist}(\mathbf{u}, \mathbf{v}) &= \sqrt{(1.0)^2 + (-1.0)^2 + (-2.0)^2 + (-1.0)^2} \\ &= \sqrt{1.0 + 1.0 + 4.0 + 1.0} \\ &= \sqrt{7} \approx 2.65\end{aligned}$$

Euclidean Distance - Example

Unit 1

$$\mathbf{v} - \mathbf{w} = [2.0, -3.0, 1.0, 2.0]$$

$$\begin{aligned}\text{dist}(\mathbf{v}, \mathbf{w}) &= \sqrt{(2.0)^2 + (-3.0)^2 + (1.0)^2 + (2.0)^2} \\ &= \sqrt{4.0 + 9.0 + 1.0 + 4.0} \\ &= \sqrt{18} \approx 4.24\end{aligned}$$

$$\mathbf{u} - \mathbf{w} = [3.0, -4.0, -1.0, 1.0]$$

$$\begin{aligned}\text{dist}(\mathbf{u}, \mathbf{w}) &= \sqrt{(3.0)^2 + (-4.0)^2 + (-1.0)^2 + (1.0)^2} \\ &= \sqrt{9.0 + 16.0 + 1.0 + 1.0} \\ &= \sqrt{27} \approx 5.20\end{aligned}$$

For Regression (Predicting a Number) - Find the Mean.

- **Measure:** Calculate the Euclidean distance between the new query point and *every* point in the training dataset.
- **Select:** Identify the K training points with the smallest distances (the nearest neighbors).
- **Predict:** Average the *target values* (not the distances!) of these K neighbors to approximate the query's unknown value.

Note: Measure and Select are both exhaustive operations.

Euclidean distance is the linear-algebraic notion of similarity. KNN turns this idea of geometric proximity into an estimate of an unknown function assuming that nearby points tend to have similar target values.

Linear algebra supplies the geometry, and machine learning supplies the statistical assumption that *nearby points or neighbors behave similarly*.

1. The Distance Between Query and Data Point

Let $\mathbf{x}_q$ be a new query data point, and $\mathbf{x}_i$ be a training data point in an n -dimensional feature space.

The Euclidean distance between these two points is:

$$\text{dist}(\mathbf{x}_q, \mathbf{x}_i) = \sqrt{\sum_{j=1}^n (x_q^{(j)} - x_i^{(j)})^2}$$

Let $\mathbf{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_M, y_M)\}$ be the training dataset, where y_i is the *known scalar target value*.

For a given query point $\mathbf{x}_q$, let $N_K(\mathbf{x}_q) \subset \mathbf{D}$ represent the subset containing its K -closest neighbors.

2. The K-NN Regression Prediction Formula

Recall D is the training dataset and $N_K(\mathbf{x}_q)$ is the set of K -nearest neighbors.

The predicted value $\hat{y}(\mathbf{x}_q)$ is computed as the arithmetic mean of the target values of those neighbors:

$$\hat{y}(\mathbf{x}_q) = \frac{1}{K} \sum_{(\mathbf{x}_i, y_i) \in N_K(\mathbf{x}_q)} y_i$$

Note: $\hat{y}$ is an approximation of the true unknown value, subject to the neighborhood definition:

$$N_K(\mathbf{x}_q) = \arg \min_{\substack{S \subset D \\ |S|=K}} \sum_{(\mathbf{x}_i, y_i) \in S} \text{dist}(\mathbf{x}_q, \mathbf{x}_i)$$

$N_K(\mathbf{x}_q)$ is the set of K training samples nearest to $\mathbf{x}_q$.

$$N_K(\mathbf{x}_q) = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K\}$$

where $\text{dist}(\mathbf{x}_q, \mathbf{x}_1) \leq \dots \leq \text{dist}(\mathbf{x}_q, \mathbf{x}_K) \leq \text{dist}(\mathbf{x}_q, \mathbf{x}_{K+1})$

Euclidean distance is the linear-algebraic notion of similarity. KNN turns this idea of geometric proximity into an estimate of an unknown function assuming that nearby points tend to have similar target values.

Linear algebra supplies the geometry, and machine learning supplies the statistical assumption that *nearby points or neighbors behave similarly*.

- When the entries of a vector represent different types of quantities, we should carefully choose the units used to represent the numerical values of the entries.
- Different components/entries should have approximately equal status in determining the vector distance.
- The numerical values should be approximately of the same magnitude.

For any vector $\mathbf{x} \in \mathbb{R}^n$, the *mean* of its entries is

$$\text{avg}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n x_i$$

and the associated *de-meaned* vector is

$$\bar{\mathbf{x}} = \mathbf{x} - \text{avg}(\mathbf{x})\mathbf{1}_n$$

Notice $\text{avg}(\bar{\mathbf{x}}) = 0$, that is, the mean value of the entries of $\bar{\mathbf{x}}$ is zero.

What is $\bar{x}$ if all entries of $\mathbf{x}$ are equal?

The *standard deviation* of an n -vector $\mathbf{x}$ is defined as the RMS value of the de-meaned vector $\mathbf{x} - \text{avg}(\mathbf{x})\mathbf{1}_n$:

$$\begin{aligned}\text{std}(\mathbf{x}) &= \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \text{avg}(\mathbf{x}))^2} \\ &= \sqrt{\frac{(x_1 - \text{avg}(\mathbf{x}))^2 + \cdots + (x_n - \text{avg}(\mathbf{x}))^2}{n}} \\ &= \frac{\|\bar{\mathbf{x}}\|}{\sqrt{n}} \\ &= \frac{\|\mathbf{x} - (\mathbf{1}^T \mathbf{x} / n) \mathbf{1}\|}{\sqrt{n}}\end{aligned}$$

- The standard deviation of a vector is **zero** only when all its entries are equal.
- The standard deviation of a vector is *small* when the entries of the vector are nearly the same.

The standard deviation of a vector $\mathbf{x}$ tells us the typical amount by which its entries deviate from their average value.

For most choices of $\mathbf{b}$, however, there is no n -vector $\mathbf{x}$ for which $\mathbf{Ax} = \mathbf{b}$. As a compromise, we seek an $\mathbf{x}$ for which $\mathbf{r} = \mathbf{Ax} - \mathbf{b}$, which we call the *residual* (for the equations $\mathbf{Ax} - \mathbf{b}$), is as small as possible.

We should choose $\mathbf{x}$ so as to minimize the norm of the residual, $\mathbf{Ax} - \mathbf{b}$.

Minimizing the norm of the residual and its square are the same, so we can just as well minimize

$$\|Ax - b\|^2 = \|r\|^2 = r_1^2 + \cdots + r_m^2,$$

the sum of squares of the residuals.